Aaditya Yadav

Phone: +918356935532
Email: aaditya.yadav2024@nst.rishihood.edu.in

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2024-2028	Bachelor of Engineering (Computer Science and AI)	Newton School of Technology, Rishihood University	7.03/10.0
2022-2023	Intermediate (Class XII)	Narayana e-Techno School and Jr College	7.067/10.0
2020-2021	Matriculation (Class X)	Rita Memorial High School	8.3/10.0

PROJECTS

EduPulse — AI-Powered Academic Coaching Platform (Github) (Demo) April 2026

- Built an AI-driven academic intelligence platform that clusters students into performance tiers using **K-Means** and predicts **pass probability** via **Gradient Boosting**, enabling data-driven academic performance analysis and insights.
- Developed an autonomous **LangGraph** tutor agent with **RAG** (ChromaDB + Hugging Face embeddings) for context aware resource retrieval and stateful tutoring sessions powered by **Groq** (LLAMA 3.3-70B) with memory management.
- Implemented a student **performance prediction** workflow combining **clustering** and **supervised learning** to support performance segmentation, pass-probability estimation, and data-driven academic insights for better decision-making.
- **Tech Stack:** Python, **Streamlit**, Scikit-learn, **LangGraph**, **LangChain**, Groq API, ChromaDB, Pandas, Matplotlib.

Nexus Research: Agentic AI Assistant for Reseach (Github) (Demo) March 2026

- Built a dual-mode research analysis platform combining a **Classical NLP pipeline** (TF-IDF, LDA topic modeling, extractive summarization) with **LangGraph**-powered AI agent for intelligent document processing and research insights.
- Integrated live web search, multi-source retrieval to generate Whole LLM based reports using **Groq (LLAMA 3.3-70B)**.
- Designed an end-to-end document-processing and retrieval workflow to extract relevant information from diverse research sources and organize it into structured, **LLM**-generated reports for efficient **research synthesis** and knowledge discovery.
- **Tech Stack:** Python, **Streamlit**, NLTK, spaCy, **Gensim** (LDA), Scikitlearn, LangGraph, BeautifulSoup4, GroqAPI.

SKILLS

- **APIs & Testing:** API, API testing, Postman, OAuth 2.0
- **Cloud & DevOps:** AWS, Git and Github, Docker, Kubernetes
- **Computer Language:** C++, Python, TypeScript, JavaScript, SQL
- **AI / ML:** PyTorch, Machine Learning, GenAI, Transformers & Generative Models, LLM, Multimodal LLMs, LangChain, LangGraph, Natural Language Processing, LangSmith, Scikit-learn, Deep Learning (DL)
- **Databases & Data Tools:** MySQL, NoSQL, PostgreSQL, MongoDB, Tableau, Excel, Spreadsheet
- **Frameworks & Libraries:** Pandas, NumPy, Matplotlib & seaborn, Prisma ORM, React, Node.js, Express JS, SciPy

RELEVANT COURSEWORK

- **Data Structures & Algorithms:** solve graphs trees and DP problems with optimal time/ space complexity.
- **Generative AI:** Built RAG powered LLM apps with embeddings and FAISS; structured output curb hallucinations.
- **Advanced Machine Learning:** Optimized scikit-learn ensembles (random forest, boosting) via cross-validation.
- **Deep Learning:** Trained TensorFlow CNNs using Adam, dropout and L2 regularizatiobns to curb overfitting.
- **Data & Visual Analytics:** Analyzed Large datasets with Pandas , EDA and Tableau dashboards to surface insights.

ACHIEVEMENTS

WorldQuant BRAIN — International Quant Championship (IQC) 2026

- Competed in a multi-stage international quantitative finance competition, securing **global ranks of 12,677 in Stage 2 and 12,819 in Stage 1**, applying quantitative problem-solving and analytical reasoning to competition challenges.
- Advanced through the competition stages, building practical exposure to **quantitative finance concepts**, **alpha research**, and **systematic investment strategies**.
- **Certificate Link**: (Certificate)